Seatwork 6.1 | Practicing with JavaScript Operators	
Course Code: CPE025 B	Program: BSCPE
Course Title: Software Design	Date Performed: 02/26/2026
Section: CPE23S4	Date Submitted: 02/26/2026
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3. Procedure	

3.0.10 SECTION PRACTICE

Arithmetic operators

Exercise 1: Fill in the missing operators to get the expected result (replace the underscore symbol with the appropriate operator):

```
console.log(2 _ 3 _ 1);      // expected 7
console.log(2 _ 4);       // expected 16
console.log(5 _ 1);       // expected 5
console.log(8 _ 2 _ 5 _ 2); // expected 39
```

Sample Solution

Comparision operators

Exercise 2: Fill in the missing comparison operators in that such a way that all expressions result in `true` (replace the underscore symbol with the appropriate operator):

```
console.log(4 * 5 _ 20);
console.log(6 * 5 _ "30");
console.log(-17 _ 0);
console.log(25 _ 1);
console.log(2 + 2 * 2 _ 4);
```

Sample Solution

Logical operators

Exercise 3: Fill in the missing comparison operators in that such a way that all expressions result in `true` (replace the underscore symbol with the appropriate operator):

```
console.log(true _ false);
console.log(false _ false);
console.log(false _ false _ true);
console.log(true _ false _ false && true);
```

Sample Solution

3.2.4 SECTION PRACTICE

Exercise: Using everything you've learned up until this point, write a script that asks a user about the width, height, and length of a box, then calculate and return to the user the volume of this box.

As an example, a box with `width = 20`, `height = 10`, and `length = 50` will have a volume of 10000 (omitting units, as they are not relevant in this scenario). For now, assume that the values provided by the user are valid numbers, but if you have any ideas on how, you can try to make this script in such a way that it will be resistant to invalid values.

Sample Solution

4. Output

Arithmetic Operators

```
console.log("This is the output for Arithmetic Operators")
console.log(2 * 3 + 1);        // expected 7
console.log(2 ** 4);        // expected 16
console.log(5 * 1);        // expected 5
console.log(8 ** 2 - 5 ** 2); // expected 39
```

Figure 1. Using Arithmetic Operators

Here in figure 1 we can see different print statements which perform arithmetic operations. Based on this I observe that in javascript it allows us to do these operations without having to declare a container variable and we are able to evaluate arithmetic inside the console log statement. I also observe that this operate similar to python where + and - are for addition and subtraction while a single * is for multiplication and ** is for exponents.

```
Output

This is the output for Arithmetic Operators
7
16
5
39

--- Code Execution Successful ---
```

Figure 2. Output

Based on this output I can say the program has the correct operators.

Comparison Operators

```
console.log("This is the output for Comparison Operators") // all must return TRUE
console.log(4 * 5 === 20);
console.log(6 * 5 =="30");
console.log(-17 < 0);
console.log(25 > 1);
console.log(2 + 2 * 2 != 4);
```

Figure 3. Using Comparison Operators

In the code block shown for figure 3 it shows the different usage of the comparison operators in order to evaluate the value and type of the initialize values involved. The first one demonstrates the strict equality operator wherein it evaluates the value and type during the comparison while the second one only evaluates the value. This is why the 30 and “30” works because in javascript this has the same value but different types. Then the other comparison is for evaluating greater than and less than and in last is the negation.

```
Output

This is the output for Comparison Operators
true
true
true
true
true
```

Figure 4. Output

Based on this output I can say the program has the correct operators.

Logical Operators

```
console.log("This is the output for Logical Operators") // all must return TRUE
console.log(true ||false);
console.log(false || ! false);
console.log(false || false || true);
console.log(true || false && false && true);
```

Figure 5. Logical Operators

In the code above it shows the logical operators utilized in javascript which is the logical or for || and logical and &&. This works similar to boolean evaluation and also returns a boolean value.

```
Output
This is the output for Logical Operators
true
true
true
true
```

Figure 6. Output

Based on this output I can say the program has the correct operators.

3.2.4 SECTION PRACTICE

```javascript
let width = prompt("Volume of the box, enter width", 0);
let height = prompt("Volume of the box, enter height", 0);
let length = prompt("Volume of the box, enter length", 0);
let volume = width * height * length;
alert(`Calculated box volume is ${volume}`);
```

Figure 7. Calculating Volume

In the figure 7. It shows the program for using the prompt to get the user input and getting the width, height and length of a box. After that it uses a variable to calculate and contain the output which is presented in an alert notification. Based on this the expected output should be in a form of prompt notification on the web.

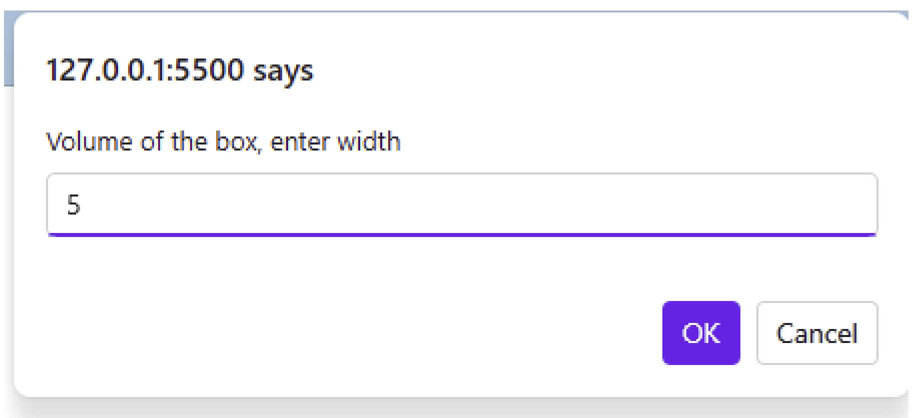

FIgure 8. Asking for Width input

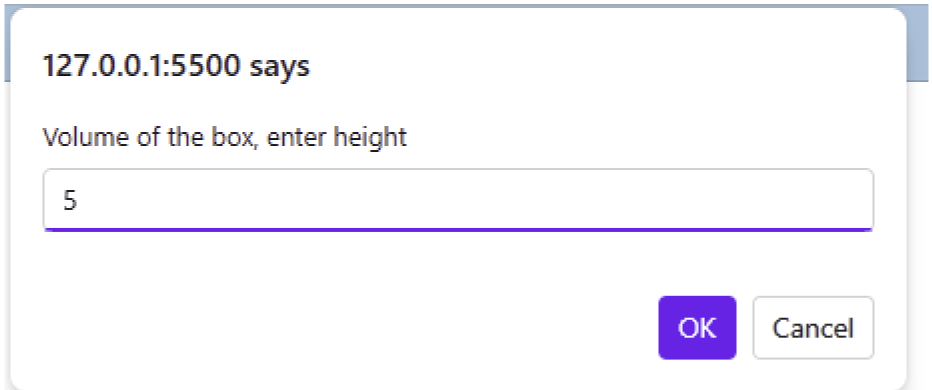

Figure 9. Asking for Height Input

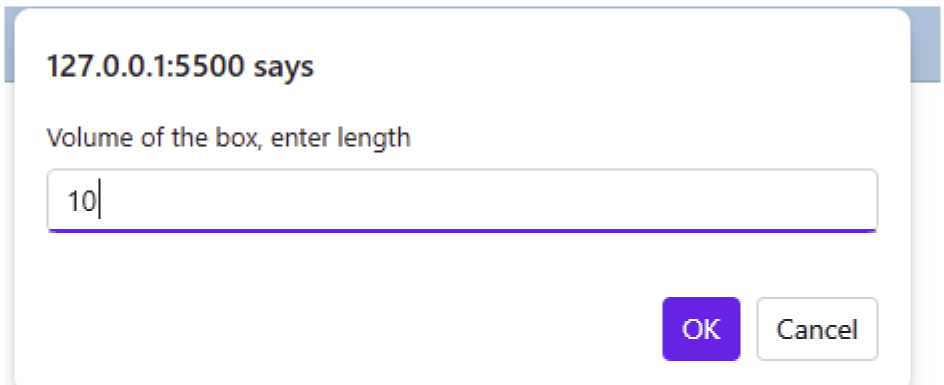

Figure 10. Asking for Length Input

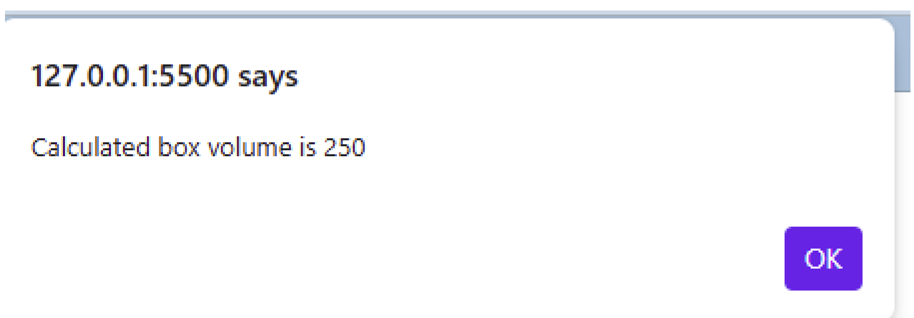

Figure 11. Output

CONCLUSION

After this activity I was able to use different operators in javascript. In this activity I utilized arithmetic operators, comparison operators, and logical operators for evaluating different values. I observe that this has almost the same rules as python with the exception for comparison operators wherein javascript has strict and loose comparison. This helps me understand different operators which I can use in evaluating variables value in javascript.